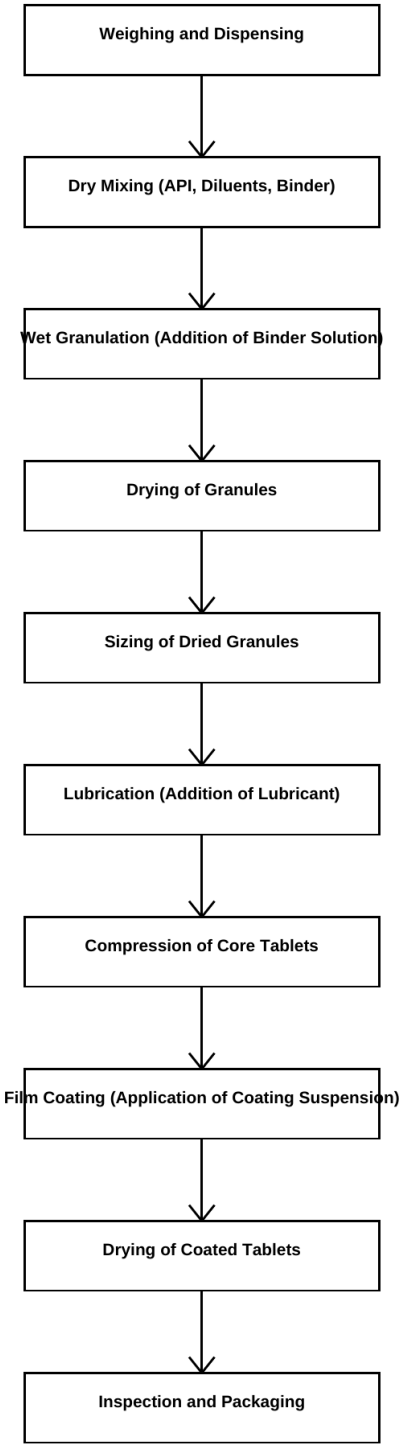
P 2.4 MANUFACTURING PROCESS DEVELOPMENT

a. Manufacturing Process Flow Chart



a. Manufacturing Process Flow

The manufacturing process for Solifenacin Succinate Tablets BP 5 mg involves several key stages, starting with the precise weighing and dispensing of all raw materials. This is followed by the dry mixing of the active pharmaceutical ingredient (API), diluents, and binder. Wet granulation is then performed by adding the binder solution. The resultant granules are dried and subsequently sized to achieve the desired particle distribution. Lubrication is carried out by adding magnesium stearate before the core tablets are compressed. Finally, the compressed tablets undergo film coating with a suspension containing Readycoat Iron Oxide Yellow, followed by drying of the coated tablets. The finished product is then inspected and packaged.

Table 1: Manufacturing Process Flow

STAGE	EQUIPMENT	CRITICAL PROCESS PARAMETER	IN PROCESS CONTROL
Weighing and Dispensing	Weighing Booths, Dispensing Area	Accuracy of weighing	Material reconciliation, visual inspection
Dry Mixing	V-Blender, Rapid Mixer Granulator (RMG)	Mixing time, speed	Uniformity of blend (in-process sampling)
Wet Granulation	Rapid Mixer Granulator (RMG)	Binder addition rate, granulation time, impeller speed	Granule size distribution, moisture content
Drying	Fluid Bed Dryer (FBD)	Inlet air temperature, drying time, air flow rate	Loss on Drying (LOD) of granules
Sizing	Multi Mill, Vibratory Sieve	Screen size, mill speed	Granule size distribution
Lubrication	V-Blender	Lubrication time, speed	Homogeneity of blend
Compression	Rotary Tablet Press	Compression force, turret speed, die fill	Tablet weight, hardness, thickness, friability
Film Coating	Coating Pan	Spray rate, atomization pressure, inlet air temperature, pan speed, coating time	Tablet appearance, weight gain, coating uniformity
Drying (Coated Tablets)	Coating Pan	Drying time, air temperature	Surface dryness
Inspection and Packaging	Inspection Belt, Blister Packing Machine	Visual inspection standards, packaging material integrity	Tablet defects, pack count, seal integrity

b. Process Description

The manufacturing process for Solifenacin Succinate Tablets BP 5 mg is a multi-stage operation designed to ensure product quality and consistency. Initially, all raw materials, including Solifenacin Succinate, Lactose Monohydrate, Maize Starch, Hypromellose, and Magnesium Stearate, are accurately weighed and dispensed according to the validated batch formula. The API, diluents, and binder (Maize Starch) are then subjected to dry mixing in a V-Blender or Rapid Mixer Granulator (RMG) to ensure homogeneity. Subsequently, wet granulation is performed in the RMG by adding a binder solution prepared with Hypromellose and Purified Water. The wet mass is then dried in a Fluid Bed Dryer (FBD) to achieve the specified Loss on Drying (LOD). The dried granules are milled to obtain a uniform particle size distribution. Following milling, Magnesium Stearate is added for lubrication in a V-Blender. The lubricated blend is then compressed into core tablets using a Rotary Tablet Press. The core tablets are subsequently film-coated in a Coating Pan using a suspension of Readycoat Iron Oxide Yellow in Purified Water. The coated tablets are dried to ensure proper film formation. Finally, the finished tablets undergo visual inspection for any defects and are then packaged into ALU/ALU blister packs.

c. Critical Process Parameters

Critical Process Parameters (CPPs) have been identified and controlled throughout the manufacturing process to ensure the consistent quality of Solifenacin Succinate Tablets BP 5 mg. These parameters, along with their associated in-process controls, are detailed in the following table. The development and validation of the process have confirmed that controlling these CPPs within defined ranges ensures that the critical quality attributes (CQAs) of the final product, such as assay, dissolution, uniformity of dosage units, and impurity profile, are met.

Table 2: Critical Process Parameters

STAGE	EQUIPMENT	CRITICAL PROCESS PARAMETER	IN PROCESS CONTROL
Dry Mixing	V-Blender, RMG	Mixing time, speed	Uniformity of blend (in-process sampling)
Wet Granulation	RMG	Binder addition rate, granulation time, impeller speed	Granule size distribution, moisture content
Drying	FBD	Inlet air temperature, drying time, air flow rate	Loss on Drying (LOD) of granules
Sizing	Multi Mill	Screen size, mill speed	Granule size distribution
Lubrication	V-Blender	Lubrication time, speed	Homogeneity of blend
Compression	Rotary Tablet Press	Compression force, turret speed, die fill	Tablet weight, hardness, thickness, friability
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